Euclidean Geometry In Mathematical Olympiads

Russian Olympiad 1996 Problem 1.41

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Solution

Let $\angle BAE = \angle CDF = x$, $\angle EAF = \angle FDE = y$, and $\angle FAD = z$. Since $\angle EAF = \angle FDE$, we have that $EFDA$ is cyclic, so $\angle FED = z$. Then, from $\triangle EFD$, we have
$\angle EFD = 180 - (y+z) \rightarrow \angle CFD = y+z \rightarrow \angle FCD = \angle C = 180 - (x+y+z)$.
Since $\angle A = x+y+z$, we have $\angle A + \angle C = 180$, so $ABCD$ is cyclic. Then, $\angle BAC = \angle CDB$, and since $\angle BAF = \angle CDE$, we have $\angle FAC = \angle EDB$.

This post has been edited 2 times. Last edited by kootrapali, Feb 29, 2020, 7:06 PM

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Aug 12, 2020, 8:49 PM • 1 #9

Solution

First note $EFDA$ is cyclic since $\angle EAF = \angle EDF$. Now let $\angle EAD = x$ and $\angle BAE = y = \angle FDC$ (this is given). Now we know that $EAFD$ is cyclic so $\angle EFD = 180 - \angle EAD = 180 - x \implies \angle DFC = x \implies FCD = 180 - x - y$.
Now $\angle BAD = x + y$ so the opposite angles of quadrilateral $ABCD$ sum to 180° and thus $ABCD$ is cyclic. Now $\angle CAB = \angle BDC$ by cyclic $ABCD$. Now $\angle CAB = y + \angle EAF + \angle FAC$ and $\angle CDB = y + \angle FDE + \angle EDB$. We have $\angle cAB = \angle CDB$ so
$y + \angle EAF + \angle FAC = y + \angle FDE + \angle EDB \implies \angle EAF + \angle FAC = \angle FDE + \angle EDB$.
Finally $\angle EAF = \angle EDF$ so we have $\angle FAC = \angle EDB$ as desired.

Solution with directed angles

Claim : ABCD is cyclic

Proof:

First note that $\measuredangle EAF = \measuredangle EDF$ so we have $EADF$ cyclic. Now because of this we know $\measuredangle EAD = \measuredangle EFD$. Now to prove the claim we simply prove $\measuredangle BAD = \measuredangle BCD$. Note $\measuredangle = \measuredangle BAE + \measuredangle EAD$.

Now consider $\triangle FDC$. Note that by triangle sum we have
$\measuredangle FCD + \measuredangle CDF + \measuredangle DFC = 0 \implies \measuredangle FCD = \measuredangle BCD = -(\measuredangle CDF + \measuredangle DFC) = \measuredangle FDC + \measuredangle CFD$.
We have $\measuredangle BAE = \measuredangle FDC$ and $\measuredangle CFD = \measuredangle EFD = \measuredangle EAD$ so $\measuredangle BCD = \measuredangle EAB + \measuredangle EAD = \measuredangle BAD$.

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